

isrc_ac.ebe

Attributes

```
ebe name=isrc_ac
Jacobian: constant
nodes: p n
rparms:
+ Im=1
+ f_hz=1
+ phi_deg=0
+ t0=0
+ idc=0
outparms: i v pwr
outparms_ac: i_ac v_ac S
```

Description

`isrc_ac.ebe` is an ac current source. The real parameters `Im`, `f_hz`, `phi_deg`, `t0`, and `idc` represent I_m , f , ϕ , t_0 , and I_{dc} respectively, in the following equation for the source current:

$$I_s(t) = I_m \sin(2\pi f(t - t_0) + \phi) + I_{dc} .$$

Note that the phase angle needs to be supplied in degrees. It is internally converted to radians. This current flows from the external circuit to node `n`, and then to node `p`. The branch current (from `n` to `p`), the branch voltage ($v_p - v_n$), and the power $p = (v_p - v_n) \times i$ are made available as output variables, `i`, `v`, `pwr`, respectively.

AC model is also implemented, and the branch current, branch voltage, and complex power are made available as AC output variables, `i_ac`, `v_ac`, and `S`, respectively.